Question 6: Java program to count the number of words using Hadoop MapReduce

Hadoop MapReduce

Aim

To count the occurrences of each word in a text file using Hadoop MapReduce.

Procedure

1. Prepare a text file containing sample sentences and upload it to HDFS.

2. The mapper converts each line to lowercase, splits it into words and emits (word, 1).

3. The framework groups values by word.

4. The reducer adds all values for each word and emits (word, total count).

5. Compile, package and run the program; display the reducer output from HDFS.

Program

```
import java.io.IOException;

import org.apache.hadoop.conf.Configuration;

import org.apache.hadoop.fs.Path;

import org.apache.hadoop.io.*;

import org.apache.hadoop.mapreduce.*;

import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;

import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class WordCount {

public static class M extends Mapper<LongWritable,Text,Text,IntWritable> {

private static final IntWritable ONE = new IntWritable(1);

public void map(LongWritable key, Text value, Context c)

throws IOException, InterruptedException {

String line = value.toString().toLowerCase();

for (String word : line.split("\\W+"))

if (!word.isEmpty()) c.write(new Text(word), ONE);

}

}

public static class R extends Reducer<Text,IntWritable,Text,IntWritable> {
```

```
public void reduce(Text word, Iterable<IntWritable> values, Context c)
throws IOException, InterruptedException {
int sum = 0;
for (IntWritable x : values) sum += x.get();
c.write(word, new IntWritable(sum));
}
}
public static void main(String[] args) throws Exception {
Job job = Job.getInstance(new Configuration(), "Word Count");
job.setJarByClass(WordCount.class);
job.setMapperClass(M.class);
job.setReducerClass(R.class);
job.setOutputKeyClass(Text.class);
job.setOutputValueClass(IntWritable.class);
FileInputFormat.addInputPath(job, new Path(args[0]));
FileOutputFormat.setOutputPath(job, new Path(args[1]));
System.exit(job.waitForCompletion(true) ? 0 : 1);
}
}
```

Compile and execute

```
export HADOOP_CLASSPATH=$(hadoop classpath)
javac -classpath "$HADOOP_CLASSPATH" -d classes WordCount.java
jar -cvf wordcount.jar -C classes .
hdfs dfs -mkdir -p /input/wc
hdfs dfs -put input.txt /input/wc/
hadoop jar wordcount.jar WordCount /input/wc /output/wc
hdfs dfs -cat /output/wc/part-r-00000
```

Output of Program

```
bigdata 2
hadoop 2
hello 1
```

world 1

Result

The Hadoop MapReduce job successfully calculates the frequency of every word in the input file.